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$K_2TeP_2O_8$: a new telluro-phosphate with a pentagonal Te–P–O layer structure†

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A new telluro-phosphate $K_2TeP_2O_8$ was synthesized by a high temperature solution method. It crystallizes in the orthorhombic space group $P2_12_12$ and features a $[TeP_2O_8]^{2-}$ layer structure, which is the first layer structure built by Te–O and P–O groups among the telluro-phosphates. More interestingly, the layer structure in $K_2TeP_2O_8$ is nearly pentagonal. First-principles theoretical calculations suggest that $K_2TeP_2O_8$ possesses a large birefringence compared to phosphates. Furthermore, IR spectra, UV-vis-NIR diffuse reflectance spectroscopy data, thermal behavior, and electronic structure calculations were studied.

1 Introduction

Tellurites and tellurates, used as nonlinear optical,[1–5] ferroelectric,[6,7] magnetic,[8,9] and thermoelectric materials,[10] have received considerable attention for their wide applications due to their fascinating structural chemistry. And the rich structural chemistry can be attributed to the variable coordination environment of the TeO_x (x = 3, 4, 5, 6) groups, such as trigonal pyramidal $(TeO_3)^{2-}$, seesaw $(TeO_4)^{4-}$, square pyramidal $(TeO_5)^{6-}$, and octahedral $(TeO_6)^{6-}$. Similarly, borates can ordinarily be three- or four-coordinated and have attracted great attention from our group over the past few years.[11–18] While in phosphates, although the P atoms can solely be four-coordinated under ordinary pressure, the P–O units can not only form isolated PO_4 or P_2O_7 groups,[19–22] but also build P_nO_{3n+1} clusters,[23,24] P_nO_{3n} cyclic rings or infinite $(PO_3)_\infty$ chains.[25–30] Therefore, to obtain a richer structural variety, TeO_x groups could be combined with the $(PO_4)^{3-}$ groups. As a result, a number of telluro-phosphates have been reported,[31–39] such as $A_3B_3TeP_2O_{14}$ (A = Sr, Ba, or Pb; B = Mg or Zn),[31,32] $BaTeMO_4(PO_4)$ (M = Nb, Ta),[33] $M_2TeMo_2O_6(PO_4)_2$ (M = K, Rb, Cs, Tl),[34] $Ba_2TeO(PO_4)_2$[35] and $Te_2O(PO_4)_2$.[36]

Among the inorganic telluro-phosphates (including composite telluro-phosphates), the Te atoms are six-coordinated in most structures (39 of total 55) based on the inorganic crystal structure database (ICSD) (2018.3). And the proportion of non-centrosymmetrical structures with the TeO_6 units is 38%, which is twice that with TeO_n (n = 3, 4 and 5). When the Te atoms are six-coordinated, the TeO_6 octahedra are isolated to the PO_4 tetrahedra. While the Te atoms are three-, four- or five-coordinated, besides being isolated to the PO_4 groups, the TeO_x (x = 3, 4, 5) groups can be connected with the PO_4 groups to form 3D frameworks or 1D chains. However, no 2D layer has been built by Te–O and P–O groups to date in telluro-phosphates. Here we report a new telluro-phosphate, $K_2TeP_2O_8$, with a layer structure constituted of TeO_5 and PO_4 groups. It is quite rare that $K_2TeP_2O_8$ features a nearly pentagonal Te–P–O layer structure (Fig. 1). Moreover, a large birefringence of $K_2TeP_2O_8$ is obtained through first-principles theoretical calculations, which may be attributed to its Te–P–O layer structure.

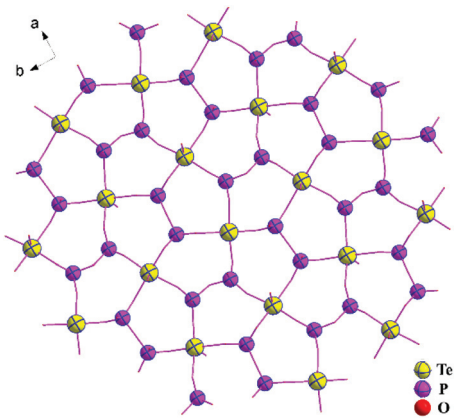

Fig. 1 The pentagonal layer structure constructed from TeO_5 and PO_4 groups.

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2 Experimental methodology

2.1 Synthesis

Single crystals of $K_2TeP_2O_8$ were grown through a high temperature solution method. The mixtures of stoichiometric amounts of K_2CO_3 (1 mmol, 0.138 g), TeO_2 (1 mmol, 0.160 g) and $NH_4H_2PO_4$ (2 mmol, 0.230 g) were put into a platinum crucible, heated up to 750 °C and held at this temperature for 12 h. Then the mixtures were cooled to 500 °C at a rate of 3 °C h^{-1} and cooled to room temperature by switching off the furnace. A transparent block shaped crystal with dimensions of 0.114 × 0.095 × 0.093 mm^3 was selected for single crystal structure determination.

Polycrystalline powder samples of $K_2TeP_2O_8$ were synthesized by a traditional solid state reaction with stoichiometric amounts of K_2CO_3, TeO_2 and $NH_4H_2PO_4$. The mixtures were ground, presintered at 450 °C for 12 h and heated to its sintering temperature of 550 °C. Pure powder samples of $K_2TeP_2O_8$ were obtained by maintaining this temperature for five days with several intermittent grindings.

2.2 Structure determination

A single crystal with a size of 0.114 × 0.095 × 0.093 mm^3 was selected for the single crystal X-ray measurements. The data were collected on a Bruker SMART APEX II CCD diffractometer using monochromatic Mo Kα radiation (λ = 0.71073 Å) and integrated with the SAINT program.[40] The calculations were completed with the SHELXTL crystallographic software package programs.[41] SHELXS-97 and PLATON were chosen to determine the structure and check the symmetry elements, respectively.[42] Table 1 shows the crystal parameters and structure refinements. And Table S1† includes the atomic coordinates and equivalent isotropic displacement parameters.

Table 1 Crystal data and structure refinements for $K_2TeP_2O_8$

Empirical formula	$K_2TeP_2O_8$
Temperature	296(2) K
Wavelength	0.71073 Å
Crystal system	Orthorhombic
Space group	$P2_12_12$
Formula weight	395.74
a (Å)	11.9684(5)
b (Å)	12.1871(5)
c (Å)	5.5283(2)
Z, volume (Å^3)	4, 806.36(6)
ρ_{Calcd} (Mg m^{-3})	5.129
μ (mm)	9.560
F(000)	736
R(int)	0.0216
Goodness-of-fit on F^2	0.983
Final R indices [$F_o^2 > 2\sigma(F_o^2)$]^a	R_1 = 0.0306, wR_2 = 0.0701
R indices (all data)^a	R_1 = 0.0315, wR_2 = 0.0707
Absolute structure parameter	0.01(6)
Largest diff. peak and hole (e Å^{-3})	1.341 and −1.584

^{a} $R_1 = \sum||F_o| - |F_c||/\sum|F_o|$ and $wR_2 = [\sum w(F_o^2 - F_c^2)^2/\sum wF_o^4]^{1/2}$ for $F_o^2 > 2\sigma(F_o^2)$.

2.3 Powder X-ray diffraction

Powder X-ray diffraction patterns (XRD) of $K_2TeP_2O_8$ were obtained on an automated Bruker D2 X-ray diffractometer using Cu Kα radiation (λ = 1.5418 Å). The samples were scanned with a scan step width of 0.02° and a step time of 1 s over the range of 10°–70° (2θ).

2.4 Infrared and UV-vis-NIR diffuse reflectance spectroscopy

Using a Shimadzu IR Affinity-1 Fourier transform infrared spectrometer, the infrared spectrum of $K_2TeP_2O_8$ was recorded in the frequency region from 400 to 4000 cm^{-1}.

UV-vis-NIR diffuse reflectance data for polycrystalline powders of $K_2TeP_2O_8$ were collected with a Shimadzu SolidSpec-3700DUV UV-VIS-IR spectrophotometer in the range from 190 to 2600 nm.

2.5 Thermal analysis

The thermal gravity-differential scanning calorimetric (TG-DSC) analysis was conducted on a NETZSCH STA 449C simultaneous thermal analyzer. At a heating rate of 10 °C min^{-1}, the sample was heated to 600 °C under a constant flow of nitrogen gas.

2.6 Second-harmonic generation test

Powder SHG measurement of $K_2TeP_2O_8$ was carried out using a Nd:YAG pulsed solid-state laser (λ = 1064 nm, 10 kHz, 10 ns).

2.7 Computational methods

The electronic structures and optical property calculations of the title compounds were performed by employing the plane-wave pseudopotential method of the CASTEP module.[43] Norm-conserving pseudopotentials (NCP)[44,45] were chosen. The valence electrons were set as: O $2s^22p^4$, P $3s^23p^3$, Te $5s^25p^4$, and K $3s^23p^64s^1$. The exchange–correlation function was nonlocal Perdew–Burke–Ernzerhof (PBE) with the generalized gradient approximation (GGA) scheme.[46] The plane-wave energy cutoff was set at 800.0 eV. The corresponding Monkhorst–Pack k-point meshes were 2 × 2 × 4.

The SHG coefficients were calculated from the band wave functions by using the so-called length-gauge formalism derived by Aversa and Sipe[47] at a zero frequency limit. The static second-order nonlinear susceptibilities $\chi_{\alpha\beta\gamma}^{(2)}$ can be reduced as:[48]

$$\chi_{\alpha\beta\gamma}^{(2)} = \chi_{\alpha\beta\gamma}^{(2)}(\mathrm{VE}) + \chi_{\alpha\beta\gamma}^{(2)}(\mathrm{VH}), \quad (1)$$

In this sum-over-states type formalism, the total SHG coefficient $\chi^{(2)}$ is divided into contribution from the Virtual-Hole (VH), Virtual-Electron (VE) and Two-Band (TB) processes. The contribution from the TB process is zero and can be neglected.

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The formulas for calculating $\chi_{\alpha\beta\gamma}{}^{(2)}(\mathrm{VE})$ and $\chi_{\alpha\beta\gamma}{}^{(2)}(\mathrm{VH})$ are as follows:

$$\chi_{\alpha\beta\gamma}{}^{(2)}(\mathrm{VE}) = \frac{e^3}{2\hbar^2 m^3}\sum_{\mathrm{vcc'}}\int\frac{\mathrm{d}^3k}{4\pi^3}P(\alpha\beta\gamma) \times \mathrm{Im}\left[P_{\mathrm{vc}}{}^{\alpha}P_{\mathrm{cc'}}{}^{\beta}P_{\mathrm{c'v}}{}^{\gamma}\right]\left(\frac{1}{\omega_{\mathrm{cv}}{}^3\omega_{\mathrm{vc'}}{}^2}+\frac{2}{\omega_{\mathrm{vc}}{}^4\omega_{\mathrm{c'v}}}\right), \quad (2)$$

$$\chi_{\alpha\beta\gamma}{}^{(2)}(\mathrm{VH}) = \frac{e^3}{2\hbar^2 m^3}\sum_{\mathrm{vv'c}}\int\frac{\mathrm{d}^3k}{4\pi^3}P(\alpha\beta\gamma) \times \mathrm{Im}\left[P_{\mathrm{vv'}}{}^{\alpha}P_{\mathrm{v'c}}{}^{\beta}P_{\mathrm{cv}}{}^{\gamma}\right]\left(\frac{1}{\omega_{\mathrm{cv}}{}^3\omega_{\mathrm{v'c}}{}^2}+\frac{2}{\omega_{\mathrm{vc}}{}^4\omega_{\mathrm{cv'}}}\right). \quad (3)$$

Here, α, β, and γ are the Cartesian components, v and v′ denote valence bands, c and c′ refer to conduction bands, and $P(\alpha\beta\gamma)$ denotes the full permutation. The band energy difference and momentum matrix elements are denoted as $\hbar\omega_{ij}$ and $P_{ij}{}^{\alpha}$, respectively.

3 Results and discussion

3.1 Structure

$K_2TeP_2O_8$ crystallizes in the orthorhombic space group $P2_12_12$. There are one unique Te site, two unique P sites, two unique K sites and nine unique O sites in the asymmetric unit. In the structure, the Te atoms are five-coordinated, while the P atoms are four-coordinated and two PO_4 tetrahedra constitute the P_2O_7 unit. The isolated TeO_5 units and P_2O_7 units share the vertexes to form the layer structure (Fig. 2a). The K atoms are nine-coordinated and reside between the layers (Fig. 2b). As shown in Fig. 2c, the TeO_5 groups are nearly placed in the opposite direction, which is consistent with the result of its dipole moments ($K_2TeP_2O_8$ crystallizes in the nonpolar point group 222 and its dipole moments are cancelled). The Te–O and P–O bond lengths in $K_2TeP_2O_8$ range from 1.779(6) to 2.325(8) Å and 1.473(7) to 1.630(12) Å, respectively (Table S2†). And the bond distances of K–O vary from 2.621(7) to 3.299(12) Å. The O(9) atoms, which connect with the $P(1)O_4$ tetrahedra to build the $P(1)_2O_7$ units, are split into two positions (Fig. S1†). The calculated valence states in $K_2TeP_2O_8$ are in agreement with the expected oxidation states (Te, 4.10; P, 4.97–4.99; K, 1.00–1.12), which confirms that all the atoms adopt normal valence states.

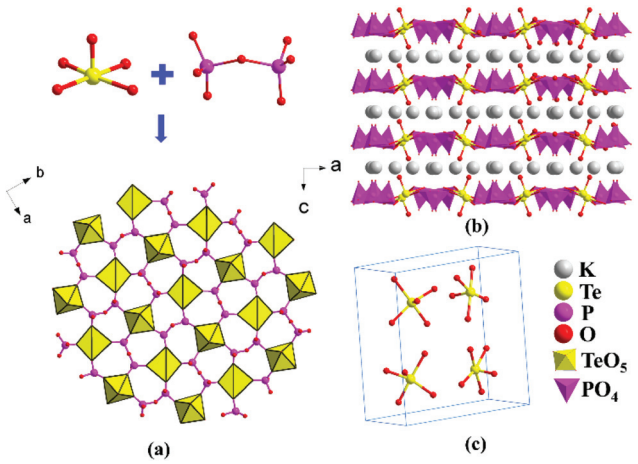

Fig. 2 The structure of $K_2TeP_2O_8$. (a) The layer structure composed of TeO_5 and P_2O_7 anion groups. (b) The 3D structure of $K_2TeP_2O_8$ along the *b* axis; K–O bonds have been omitted for clarity. (c) The arrangement of TeO_5 groups in the unit cell.

3.2 Thermal analysis

The TG-DSC curves of $K_2TeP_2O_8$ are shown in Fig. S2.† There is only one endothermic peak located at 600 °C without weight loss, which may indicate its congruently melting behaviour. To further confirm this, the powder samples were heated to 800 °C. Then the samples were cooled to 500 °C at a rate of 2 °C h^{-1}. As shown in Fig. S3,† the XRD patterns of the samples melted at 750 °C and sintered at 550 °C are both identical to those of the calculated ones. These results indicate that $K_2TeP_2O_8$ melts congruently.

3.3 Infrared spectrum

The IR spectrum of $K_2TeP_2O_8$ is shown in Fig. S4.† The asymmetric and symmetric PO_4 vibrations are observed around 1220 and 1024 cm^{-1}, respectively. The peaks located at 982 and 866 cm^{-1} are assigned to the asymmetrical stretching vibrations of the PO_4 groups. And the peaks ranging from 430 to 730 cm^{-1} mainly result from the Te–O bonds. These values are in accordance with those of other compounds containing P–O and Te–O groups.[32,49,50]

3.4 UV-vis-NIR diffuse reflectance spectroscopy

Fig. 3 shows the diffuse reflectance spectrum of $K_2TeP_2O_8$. The absorption (*K*/*S*) data are shown in the inset of Fig. 3 by the Kubelka–Munk function.[51,52] It is indicated that $K_2TeP_2O_8$ has an optical band gap of 4.6 eV.

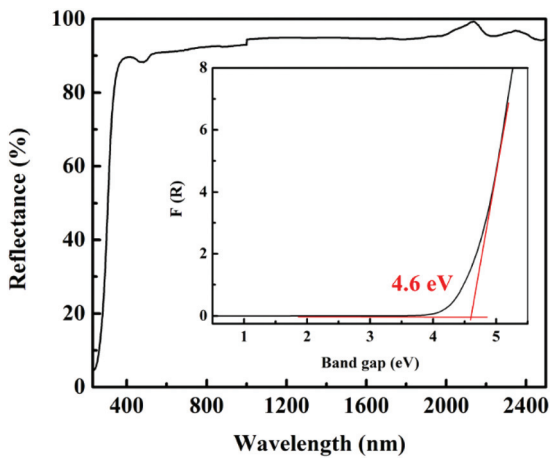

Fig. 3 UV-vis-NIR diffuse reflectance spectrum of $K_2TeP_2O_8$.

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3.5 Second-harmonic generation

The powder SHG density of $K_2TeP_2O_8$ is very weak. Although the Te^{4+} cations are stereo-chemically active, the nearly opposite arrangement of TeO_5 units (Fig. 2c) is negative to the large SHG response, and results in a weak SHG response, which is less than 0.1 KDP.

3.6 Electronic structure calculations

The band structure of $K_2TeP_2O_8$ is shown in Fig. S5,† which indicates that it is an indirect gap compound with a band gap of 4.26 eV. The calculated band gap (4.26 eV) agrees well with the experimental one (4.60 eV). In addition, the calculated SHG intensity of d_{14} is 0.25 pm V^{-1} (0.6 × KDP).

The density of states (DOS) between −15 and 15 eV is depicted in Fig. 4. In the valence band (VB) from −15 to −2.5 eV, there are O 2s and 2p orbitals, P 3s and 3p orbitals, Te 5s and 5p orbitals and K 3p orbitals. At the top of the VB, the main occupants are the O 2p orbitals with small amounts of Te 5s, 5p and P 3p orbitals while the bottom of the conduction band (CB) is dominated by the Te 5p orbitals. The high energy section of CB (above 7.5 eV from the Fermi surface) mainly consists of P 5p orbitals and K 3p and 4s orbitals. The DOS indicate that the band gap of $K_2TeP_2O_8$ is dominated by the orbital hybridization of O and Te atoms. Since the valence electrons near the Fermi level dominate the optical properties of a material, the TeO_5 groups contribute mostly to the optical properties of $K_2TeP_2O_8$.

The Te atom has a lone pair $5s^2$, and in the structure of $K_2TeP_2O_8$ it causes obvious structure distortions. So this lone pair electron is usually called stereo-chemically lone-pairs (SCALPs), which can be captured in the electronic local function (ELF, shown in Fig. S6†).[53] The ellipses, drawn in dark blue around the Te atoms are considered as the SCALPs.

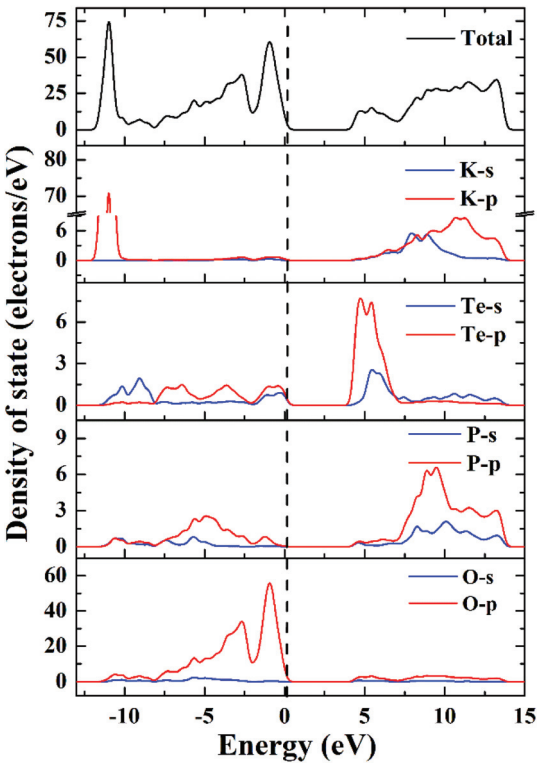

Fig. 4 DOS of $K_2TeP_2O_8$.

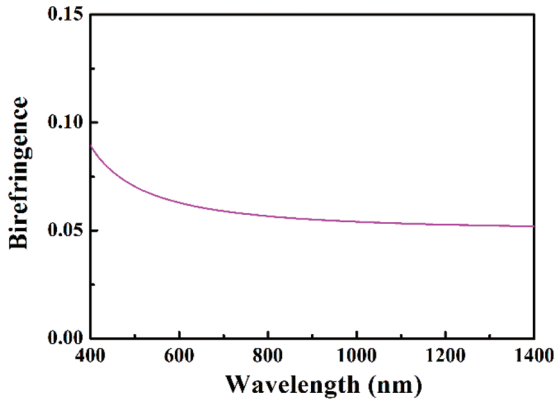

Fig. 5 Calculated birefringence of $K_2TeP_2O_8$.

The refractive indices and birefringence of $K_2TeP_2O_8$ were calculated. It shows a large birefringence more than 0.05 from 400 to 1400 nm (Fig. 5), which is larger than those of the phosphates with reported birefringence.[23,28,54–58] The values of n_x and n_y are nearly the same and obviously differ from that of n_z (shown in Table S3†), which may be due to the layer structure with large anisotropy in the directions parallel to and perpendicular to the layer. Therefore, the large birefringence of $K_2TeP_2O_8$ may be attributed to its large response electronic distribution induced by the layer structure.[59]

4 Conclusions

We have synthesized a new telluro-phosphate, $K_2TeP_2O_8$, crystallizing in a non-centrosymmetrical space group $P2_12_12$. Its most interesting structural feature is a nearly pentagonal $[TeP_2O_8]^{2-}$ layer structure. $K_2TeP_2O_8$ is a congruent melting compound according to the TG-DSC curves. And $K_2TeP_2O_8$ exhibits an experimental band gap of 4.6 eV. The DOS illustrates that the band gap is dominated by the orbital hybridization of O and Te atoms. The theoretical calculations show that $K_2TeP_2O_8$ possesses a larger birefringence compared with phosphates, which may be attributed to its layer structure.

Conflicts of interest

There are no conflicts to declare.

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